INCH-POUND

MIL-S-19500/589
30 July 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, INSULATED GATE, BIPOLAR TRANSISTOR, N-CHANNEL, SILICON TYPES 2N7366 AND 2N7367, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an insulated gate bipolar power transistor with a copackaged antiparallel ultrafast soft recovery diode, intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-258) for 2N7367 and figure 2 (TO-259) for 2N7366.

1.3 Maximum ratings. At T_C = 25°C, unless otherwise specified. 1/ 2/

Type	P_T 3/ T_C = +25°C	P_T T_A = +25°C	V_{GE}	I_{C1} 4/ T_C = +25°C	I_{C2} T_C = +100°C	I_{CM} 5/	$V_{BR(CES)}$ V_{GE} = 0 V, I_C = 1.0 mA	V_F I_F = I_{C2}
	W	W	V dc	A dc	A dc	A (pk)	V dc	V
Both	200	4	±20	45	27	220	600	1.7

Type	I_{LM} 6/	$V_{(ISO)}$ at 70K feet	$R_{\theta JC1}$ max for I_{GBT}	$V_{CE(on)}$ max, V_{GE} = 15 V, T_J = +25°C	$V_{CE(on)}$ max, I_C = I_{C2}, T_J = +150°C	T_J and T_{STG}	T_{RR} I_f = I_{C2}	$R_{\theta JC2}$ max for diode
	A		°C/W	V	V	°C	ns	°C/W
Both	220	600	0.64	3.0	2.89	-55 to +150	100	1.2

1/ Pulsed (see 4.5.1).

2/ Electrical and thermal ratings are identical for both 2N7366 and 2N7367.

3/ Derate linearly 1.56 W/°C for T_C > +25°C; $P_T = \frac{T_{JM} - T_C}{R_{\theta JC}}$.

4/ Package limited to 45 A dc.

5/ $I_{CM} = 4 \times I_C$.

6/ Clamped inductive load current.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at +25°C. 1/

Type	I_{GES} (MAX) V_{GE} = ±20 V V_{CE} = 0 V	$V_{GE(th)}$ $V_{CE} \geq V_{GE}$ I_C = 0.25 mA		I_{CES} Max V_{GE} = 0 V V_{CE} = 80% max rated V_{CE}	$V_{CE(on)}$ Max V_{GE} = 15 V $I_C = I_{C1}$	I_{RM} of diode $I_F = I_{CZ}$
	nA dc	V dc		µA dc	V	A
		Min	Max			
2N7366	100	3.0	5.5	250	3.25	10
2N7367	100	3.0	5.5	250	3.25	10

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

$V_{(ISO)}$ - Source pin-to-case isolation voltage.

$I_{(ISO)}$ - Source pin-to-case isolation voltage.

I_{LM} - Clamped inductive load current.

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3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.3.1 Lead formation and finish. Lead finish shall be solderable as specified herein. When lead formation is performed, as a minimum the vendor shall perform 100 percent hermetic seal in accordance with screen 14 and 100 percent dc testing in accordance with Group A, subgroup 2.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices shall be as followed.

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100k$, whenever bias voltage is to be applied drain-to-source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV level
1/	Gate stress test (see 4.5.3)	Gate stress test (see 4.5.3)
1/	Clamped I_L test (see 4.5.5)	Clamped I_L test (see 4.5.5)
1/	Thermal response (see 4.5.4)	Thermal response (see 4.5.4)
9 1/	Group A, subgroup 2	Group A, subgroup 2
10	MIL-STD-750, method 1042, test conditions B	MIL-STD-750, method 1042, test condition B
11	I_{GESF1}, I_{GESR1}, I_{CES1}, $V_{CE(on)}$, $V_{GE(th)}$ ΔI_{GES} = ±50 nA or 100 percent of initial value, whichever is greater. ΔI_{CES} = ±250 µA or 100 percent of initial value, whichever is greater. Subgroup 2 of table I herein.	I_{GESF1}, I_{GSER1}, I_{CES1}, $V_{CE(on)}$, $V_{GE(th)}$ Subgroup 2 of table I herein
12 2/	MIL-STD-750, method 1042, test conditions A, T_A = +125°C minimum	MIL-STD-750, method 1042, test condition A, T_A = +125°C minimum
13	Subgroups 2 and 3 of table I herein ΔI_{GES} = ±50 nA or 100 percent of initial value, whichever is greater. ΔI_{CES} = ±250 µA or 100 percent of initial value, whichever is greater. $\Delta V_{CE(ON)}$ = ±20 percent of initial value. $\Delta V_{GE(th)}$ = ±20 percent of initial value.	Subgroup 2 of table I herein ΔI_{GES} = ±50 nA or 100 percent of initial value, whichever is greater. ΔI_{CES} = ±250 µA or 100 percent of initial value, whichever is greater. $\Delta V_{CE(ON)}$ = ±20 percent of initial value. $\Delta V_{GE(th)}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.

2/ No heat sinking of devices is allowed.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. (Endpoint electrical measurements shall be in accordance with the applicable steps of table V herein.) Electrical and thermal conditions and limits are identical for both 2N7367 and 2N7366.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and table IIa and IIb herein. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 3: Condition for bond strength is condition A.

b. Subgroup 4: Condition for intermittent operation life is as follows:

Condition D, 2,000 cycles. The heating cycle shall be one minute minimum.

c. Subgroup 5: Condition for accelerated steady-state reverse bias is as follows:

Condition A, V_{CE} = 80 percent of rated, T_C = +150°C +0°C, -15°C, t = 120 hours. Read and record $V_{BR(CES)}$ (pre and post) at I_C = 1 mA. Read and record I_{CES} (pre and post), in accordance with table III.

d. Subgroup 5: Condition for accelerated steady-state gate stress is as follows:

Condition B, V_{GE} = 80 percent of rated, T_A = +175°C, t = 24 hours.

e. Subgroup 6: Condition for thermal resistance, see 4.5.2.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 3: Condition for intermittent operation life is as follows:

Condition D, 2,000 cycles. The heating cycle shall be one minute minimum.

b. Subgroup 4: Condition for bond strength is condition A.

c. Subgroup 6: Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength (tension) is test condition A, weight 10 pounds, time = 15 seconds.

b. Subgroup 6: Condition for intermittent operation life is as follows:

Condition D, 6,000 cycles. The heating cycle shall be one minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3407	V_{GE} = 0; I_C = 1.0 mA Bias condition C	$V_{BR(CES)}$	600		V
Gate to emitter voltage (threshold)	3403	$V_{CE} \geq V_{GE}$ I_C = 0.25 mA	$V_{GE(th)1}$	3.0	5.5	V dc
Gate current	3411	V_{GE} = 20 V; V_{CE} = 0 V Bias condition C	I_{GESF1}		100	nA dc
Gate current	3411	V_{GE} = -20 V; V_{CE} = 0 V Bias condition C	I_{GESR1}		-100	nA dc
Collector current	3413	Bias condition C V_{GE} = 0 V V_{CE} = 80 percent of rated V_{CE}	I_{CES1}		250	µA dc
Static collector to emitter on-state voltage	3405	V_{GE} = 15 V; I_C = I_{C2} condition A; pulsed (see 4.5.1)	$V_{CE(on)1}$		3.0	V
Static collector to emitter on-state voltage	3405	V_{GE} = 15 V; I_C = I_{C1} condition A; pulsed (see 4.5.1)	$V_{CE(on)2}$		3.25	V
Forward voltage	4011	V_G = 0 I_F = I_{C2}	V_F		1.7	V
Subgroup 3						
High temperature operation:		T_C = T_J = +125°C				
Gate current	3411	Bias condition C V_{GE} = 20 V; V_{CE} = 0 V	I_{GES}		500	nA dc
Gate current	3411	Bias condition C V_{GE} = 20 V; V_{CE} = 0 V	I_{GES}		-500	nA dc
Collector current	3413	Bias condition C V_{GE} = 0 V_{CE} = 100 percent of rated V_{CE}	I_{CES2}		5.0	mA dc
Gate to emitter voltage (threshold)	3403	$V_{CE} \geq V_{GE}$ I_C = 0.25 mA	$V_{GE(th)}$	2.0		V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Static collector to emitter on-state voltage	3405	V_{GE} = 15 V; $I_C = I_{C2}$ pulsed (see 4.5.1)	$V_{CE(on)3}$		2.85	V
Forward voltage	4011	V_G = 0 $I_C = I_{C2}$	V_F		1.5	V
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to emitter voltage (threshold)	3403	$V_{CE} \geq V_{GE}$ I_C = 0.25 mA	$V_{GE(th)}$		6.5	V dc
Subgroup 4						
Switching time test	3477	I_C = rated I_{C2} V_{GE} = 15 V dc V_{CC} = 0.8 X $V_{BR(CES)}$ R_G = 2.0Ω				
Turn-on delay time			$t_{d(on)}$		50	ns
Rise time			t_r		75	ns
Turn-off delay time			$t_{d(off)}$		300	ns
Fall time			t_f		210	ns
Total switching losses	3477	I_C = rated I_{C2} V_{GE} = 15 V dc V_{CC} = 0.8 X $V_{BR(CES)}$ R_G = 2.0Ω	E_{total}		2.8	mJ
Reverse recovery time	3473	di/dt ≤ 200 A/µs, $I_F = I_{C2}$	t_{rr}		100	ns
Maximum reverse current			I_{RM}		10	A
Subgroup 5						
Not applicable						
Subgroup 6						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471					
On-state gate charge			$Q_{g(on)}$		140	nC
Gate to emitter charge			Q_{ge}		35	nC
Gate to collector charge			Q_{gc}		70	nC

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group E inspections (all quality levels) for qualification only. 1/

Inspections	MIL-STD-750		Quality conformance inspection sample size
	Method	Conditions	
Subgroup 1			45 devices, c = 1
Thermal shock (temperature cycle)	1051	-55°C to +150°C 500 cycles	
Hermetic seal	1071		
a. Fine leak b. Gross leak			
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 2 2/			45 devices, c = 1
Steady-state reverse bias	1042	Condition A, 1,000 hours T_J = 150°C +0°C, -15°C	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7	
Steady-state gate bias	1042	Condition B, 1,000 hours T_A = +150°C	
Electrical measurements		See table V, steps 1, 2, 3, 4, 5, 6, and 7	
Subgroup 3			
DPA	2102		
Subgroup 4			
Thermal resistance	3103 3101	See 4.5.2	5 devices, c = 0
Subgroup 5			
Barometer pressure test	1001	Condition C, V_{CE} = 600 V maximum, $I_{(ISO)}$ = 0.25 mA	5 devices, c = 0

1/ In initial design verification only (1 time testing).
2/ A separate sample shall be pulled for each test.

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TABLE III. Groups A, B, C, and E electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Breakdown voltage, collector to emitter	3407	V_{GE} = 0; I_C = 1.0 mA Bias condition C	$V_{BR(CES)}$	600		V dc
2.	Gate to emitter voltage (threshold)	3403	$V_{CE} \geq V_{GE}$ I_C = 0.25 mA	$V_{GE(th)1}$	3.0	5.5	V dc
3.	Gate current	3411	V_{GE} = 20 V V_{CE} = 0 Bias condition C	I_{GES}		100	nA dc
	Gate current	3411	V_{GE} = 20 V V_{CE} = 0 Bias condition C	I_{GES}		-100	nA dc
4.	Collector current	3413	V_{GE} = 0 Bias condition C V_{CE} = 80 percent of rated V_{CE}	I_{CES}		250	µA
5.	Static collector to emitter on-state voltage	3405	V_{GE} = 15 V Condition A, pulsed (see 4.5.1) $I_C = I_{C2}$	$V_{CE(on)1}$		3.0	V
6.	Static collector to emitter on-state voltage	3405	V_{GE} = 15 V Condition A, pulsed (see 4.5.1) $I_C = I_{C1}$	$V_{CE(on)2}$		3.25	V
7.	Forward voltage drop	4011	V_G = 0 $I_F = I_{C2}$	V_F		1.7	V
8.	Thermal response	3103 3101	See 4.5.4				

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroups 3 and 4, see table III herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

b. Subgroups 5 and 6, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

Subgroups 2 and 3, see table III herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 6, see table III herein, steps 1, 2, 3, 4, 5, 6, 7, and 8.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions of pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance.

4.5.2.1 Thermal resistance for IGBT. Thermal resistance measurements shall be performed in accordance with method 3103 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ shall be 0.64°C/W. The following parameter measurements shall apply:

a.	Measuring current (I_M)	10 mA.
b.	Collector heating current (I_H)	4 A minimum.
c.	Heating time (t_H)	Steady-state (see MIL-STD-750, method 3103, for definition).
d.	Collector-emitter heating voltage (V_H)	25 V minimum.
e.	Measurement time delay (t_{MD})	30 to 60 µs.
f.	Sample window time (t_{SW})	10 µs maximum.

4.5.2.2 Thermal resistance for diode. Thermal resistance measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum limit of $R_{\theta JC(max)}$ shall be 1.2°C/W. The following parameter measurements shall apply:

a.	Measuring current (I_M)	10 mA.
b.	Collector heating current (I_H)	15 A minimum.
c.	Heating time (t_H)	Steady-state (see MIL-STD-750, method 3103, for definition).
d.	Measurement time delay (t_{MD})	30 to 60 µs.
e.	Sample window time (t_{SW})	10 µs maximum.

4.5.3 Gate stress test.

V_{GE} = 30 V minimum.

t = 250 µs minimum.

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4.5.4 Thermal response (measurements).

4.5.4.1 Thermal response (measurements) for IGBT. The ΔV_{GE} measurements shall be performed in accordance with method 3103 of MIL-STD-750. The ΔV_{GE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 3). The ΔV_{GE} measurement and conditions for each device in the qualification lot shall be submitted (read and record) in the qualification report. The chosen ΔV_{GE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. Measuring current (I_M) - - - - - - - - - - - - 10 mA.

b. Collector heating current (I_H) - - - - - - - - 4 A minimum.

c. Heating time (t_H) - - - - - - - - - - - - - - 75 mS minimum.

d. Collector-emitter heating voltage (V_H) - - - 25 V minimum.

e. Measurement time delay (t_{MD}) - - - - - - - - 30 to 60 µs.

f. Sample window time (t_{SW}) - - - - - - - - - - 10 µs maximum.

4.5.4.2 Thermal response (measurements) for diode. The ΔV_F measurements shall be performed in accordance with method 3101 of MIL-STD-750. The ΔV_{GE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 3). The ΔV_{GE} measurement and conditions for each device in the qualification lot shall be submitted (read and record) in the qualification report. The chosen ΔV_F shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. Measuring current (I_M) - - - - - - - - - - - - 10 mA.

b. Collector heating current (I_H) - - - - - - - - 50 A minimum.

c. Heating time (t_H) - - - - - - - - - - - - - - 50 mS minimum.

d. Measurement time delay (t_{MD}) - - - - - - - - 30 to 60 µs.

e. Sample window time (t_{SW}) - - - - - - - - - - 10 µs maximum.

4.5.5 Clamped inductive load. This test shall be performed in accordance with method 3490 of MIL-STD-750.

a. V_{CC} = 50 V minimum.

b. V_G = 20 V.

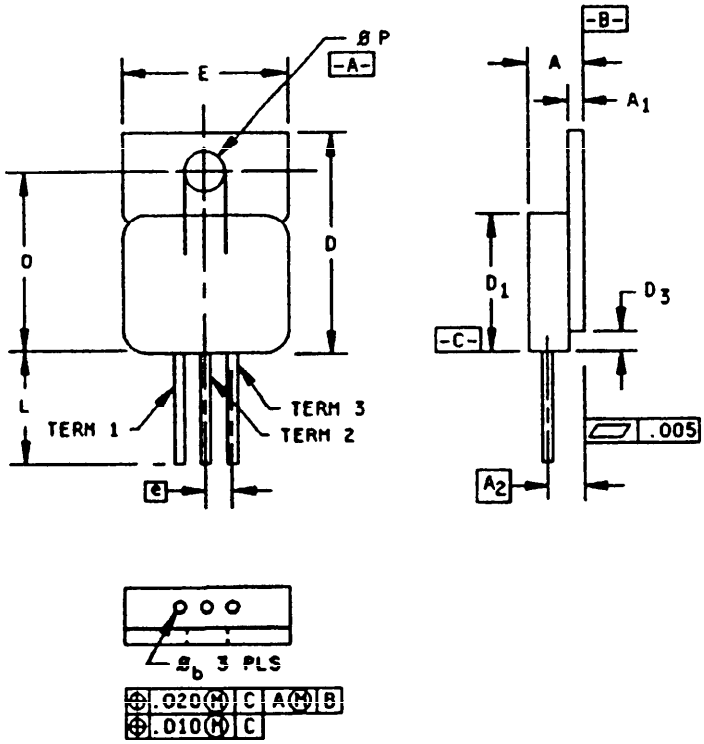
c. L = 10 µH minimum.

d. R_G = 2 ohms.

e. V_{CL} = 80 percent $V_{BR(CE)}$ maximum.

f. I_L = 220 A minimum.

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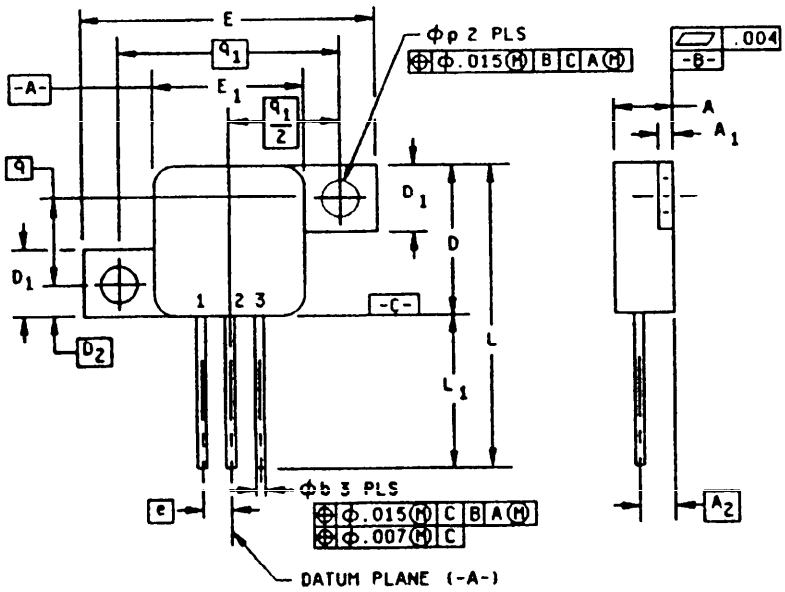


Symbol	AA	
	Min	Max
A	.240	.270
A_1	.035	.045
A_2	.140 BSC	
Øb	.055	.065
D	.815	.835
D_1	.530	.550
D_3	.000	.092
e	.200 BSC	
E	.685	.695
L	.500	.750
O	.697	.707
ØP	.155	.165

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Glass meniscus included in dimensions e and E.
4. Term 1 = Collector, Term 2 = Emitter, Term 3 = Gate.

FIGURE 1. Physical dimensions of TO-258.

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Symbol	AA	
	Min	Max
A	.240	.270
A_1	.035	.045
A_2	.140 BSC	
ϕb	.055	.065
D	.530	.550
D_1	.245	.255
D_2	.125 BSC	
e	.200 BSC	
E	1.185	1.195
E_1	.685	.695
L	1.030	1.300
L_1	.500	.750
ϕP	.120	.130
q	.285 BSC	
q_1	.940 BSC	

NOTES:
1. Refer to applicable symbol list.
2. Dimensions and tolerance in accordance with ANSI Y14.5M - 1982.
3. All dimension in inches.
4. Term 1 = Collector, Term 2 = Emitter, Term 3 = Gate.

FIGURE 2. Physical dimensions of TO-259.

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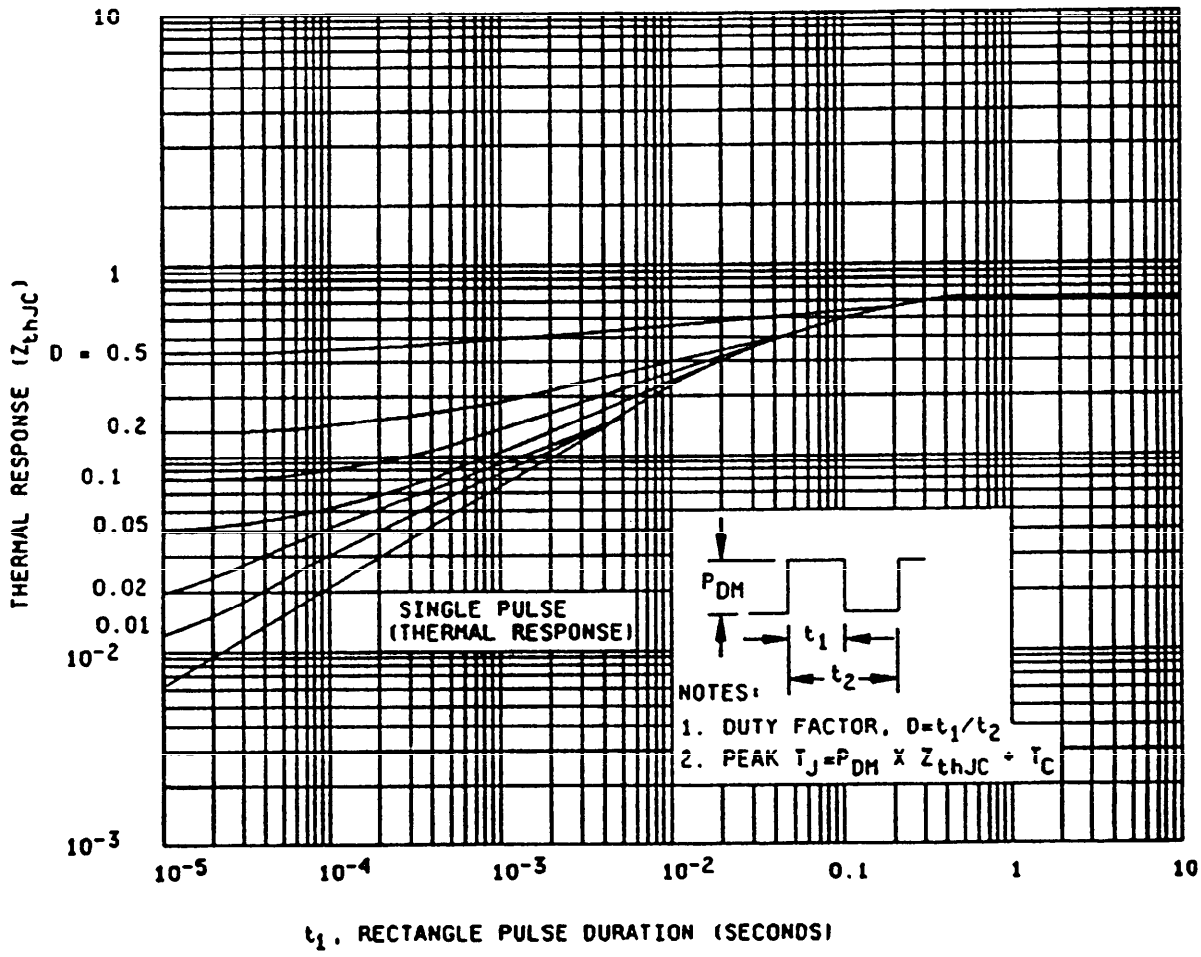


FIGURE 3. Thermal response curves.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation or lead finish may be specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN's). This information in no way implies that the manufacturer's PIN's are suitable as a substitute for the military PIN's.

Supersession and cross-reference data

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N7366	59993	IRGMVC50U
2N7367	59993	IRGMIC50U

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 70, 80, 85, 99
DLA - ES

User activity:
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1323)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/589	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, INSULATED GATE, BIPOLAR TRANSISTOR, N-CHANNEL, SILICON TYPES 2N7366 AND 2N3767, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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